

Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59 Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey

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To,
The editorial team
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We are pleased to submit our manuscript entitled “**Factors Associated with Undernutrition, Anemia and Their Coexistence Among 6-59 Months Children in Nepal: A Secondary Analysis of the Nepal Demographic and Health Survey-2022**” and supporting materials for your consideration.

We believe this paper particularly interesting because we analyzed secondary data from the nationally representative cross-sectional survey, Nepal Demographic and Health Survey 2022 (NDHS-2022) to to determine the prevalence of undernutrition, anemia, and their co-existence in Nepal, as well as determine their contributing factors among children aged 6-59 months.

We confirm that this manuscript is original and has not been published elsewhere or is under consideration for publication elsewhere. We have no conflicts of interest to disclose. Please address all correspondence concerning this manuscript to: bikram.adhikariadhitya@gmail.com

Sincerely,
Bikram Adhikari
University of Memphis
Memphis, TN, USA

Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59 Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey

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Abstract

Introduction: Undernutrition and anemia among children aged 6 to 59 months are significant public health issues in developing countries like Nepal. The co-existence of these conditions impacts childhood development. This study aimed to determine the prevalence of undernutrition and anemia, assess their co-existence, and identify contributing factors among children aged 6-59 months in Nepal.

Methods: We conducted a secondary analysis using data from 2022 Nepal Demographic and Health Survey (NDHS 2022), a nationally representative cross-sectional survey. The outcome variables were undernutrition, anemia, and their co-existence. Undernutrition was defined as the presence of stunting, wasting, underweight or any combination of these conditions. Anemia was defined as hemoglobin levels <11.0 g/dL (adjusted for altitude). We applied multivariable multinomial logistic regressions to determine factors associated with co-existence, and multivariable logistic regression to assess factors associated with undernutrition and anemia separately. We presented the results from the regression analysis using adjusted odds-ratio (aOR) and 95% confidence intervals (CI).

Results: Among 2,395 children, the weighted prevalence of undernutrition was 33.5%, anemia was 43.4%, and co-existence of undernutrition and anemia was 16.0%. Children from the richest wealth quintile, whose mothers had at least secondary education, and those whose mother participated in household decision-making had 53% (aOR: 0.47; 95% CI: 0.26 to 0.86), 48% (aOR:0.52; 95% CI: 0.32 to 0.86), and 34% (aOR: 0.66; 95% CI: 0.47 to 0.94) lower odds of experiencing the co-existence of undernutrition and anemia compared to their counterparts. Children with underweight mothers had 80% (aOR:1.80; 95% CI:1.20 to 2.70) higher odds of co-existence compared to their counterparts.

Conclusion: The prevalence of undernutrition, anemia, and their co-existence among children was high in Nepal. Interventions that improve mother's education, strengthen their roles in the family, and enhance the household financial condition are essential to reduce these conditions and improve children's nutritional status.

Keywords: *undernutrition; anemia; co-existence; 6-59-months children; Nepal; NDHS 2022*

Introduction

Undernutrition including stunting, wasting, and being underweight is a major public health challenge among children under 5 years (U-5) of age globally. This issue is particularly pronounced in developing countries like Nepal [1–4]. There are three different causes of undernutrition – immediate, underlying, and basic: The immediate causes include inadequate diet intake and repeated illnesses; The underlying causes include suboptimal child feeding and caregiver practices from mothers, lack of access to healthcare facilities, poverty, and food insecurity; basic causes include limited information, political and economic insecurity, gender inequality, and the occurrence of natural disasters [5].

Different indices of undernutrition measure different underlying development status. Stunting indicates chronic malnutrition reflecting long-term nutrient deficiencies and is linked with delayed and impaired motor and cognitive development [6]. Wasting indicates acute malnutrition associated with recent severe food shortages or illnesses leading to weight loss and is a strong predictor of mortality [6]. Underweight combines information about linear growth obstruction and weight for length or height [7].

Anemia, particularly iron deficiency anemia, is another common and pressing public health concern among children U-5 in low-, middle- and high-income countries. It results from poor nutrition and has severe adverse health consequences including impaired cognitive development, impaired immunity, disability, and an increased risk of morbidity and mortality [8–10].

In 2022, the global prevalence of stunting among children under five years of age was 22.3%, while in South Asia, it reached 31.8%. Additionally, the global prevalence of wasting was at 6.8%, with South Asia reporting a prevalence of 14.8% [11]. In 2019, global anemia prevalence was 39.8% [12]. In Nepal, the prevalence of stunting, and wasting among children under 5 was 25% and 8% respectively, and the prevalence of anemia was 43% among children 6-59 months old [13].

The co-existence of undernutrition and anemia increases the risk of childhood morbidity and mortality [14–16]. Almost fifty percent of deaths among children under the age of five are associated with undernutrition, which predominantly occurs in low- and middle-income countries (LMICs). Notably, 88% of these countries (124 out of 141) experience multiple forms of malnutrition [17]. The developmental, economic, social, and medical impacts of undernutrition and anemia among children are serious and perpetuating, for individuals and their families, communities, and countries [7].

Furthermore, anemia and undernutrition are both concentrated in socioeconomically disadvantaged groups, which share numerous multifaceted causes and complex interactions between diet, transmissible illnesses, inadequate care and unhealthy household environments that adversely affect the cognitive development and physical well-being of children [18–20]. For instance, lower mother's education level and socioeconomic status of households are linked with higher rates of malnutrition among children [18].

Though the concerns on nutrition should be explored beyond any single form of malnutrition to the co-existence of multiple forms, most studies in Nepal have focused on undernutrition and anemia separately. A few studies that have examined both conditions together primarily reported that children from poorer households, mothers with low education levels, and those with inadequate dietary diversity were more likely to experience concurrent undernutrition and anemia [21,22]. Even when both malnutrition and anemia were considered, anemia was often treated merely as a correlating factor or predictor rather than as part of a combined outcome reflecting their simultaneous occurrences. Moreover, there remains a gap in understanding how maternal factors (such as nutritional status and exposure to health information) and household-level socioeconomic characteristics interact to influence the co-existence of undernutrition and anemia among children under five in Nepal.

Therefore, this study aims to ascertain the prevalence of stunting, wasting, and underweight, as well as their co-occurrence, and the co-existence of undernutrition and anemia in Nepal. It also assesses the association of undernutrition, anemia, and co-existence of undernutrition with household economic status, maternal education and nutritional status, and exposure to televised or broadcasted health programs among children aged 6-59 months in Nepal. By identifying these factors, this study helps inform the development of targeted interventions and policies designed to mitigate malnutrition within this vulnerable population, thereby enhancing child health outcomes and alleviating the burden of malnutrition in Nepal.

Methods

Data source

In this study, we analyzed data from the Nepal Demographic and Health Survey (NDHS) conducted in 2022 [23]. NDHS 2022 is the nationally representative survey implemented by New ERA under the aegis of the Ministry of Health and Population (MoHP) with the technical support of ICF International and funding from the United States Agency for International Development (USAID) [13]. The NDHS follows standardized demographic and health survey (DHS) protocols for sampling, data collection, and quality assurance.

Ethical approval

We received permission from the official website of “the DHS program” (<https://www.dhsprogram.com>) to download and use (submitted on: 07/25/2023 and approved on 07/26/2023) NDHS 2022 dataset [23]. NDHS 2022 obtained ethical approval from the Institutional Review Board of ICF International, United States of America (Reference number: 180657.0.001.NP.DHS.01, Date: 28th April 2022). and the ethical review board of Nepal Health Research Council (Reference number: 678, Date: 30th September 2021) [13]. In the NDHS 2022, informed consent was taken from the participants before enrolling them into the study.

Study setting

This study used nationally representative data from Nepal, a landlocked country located in Southeast Asia with an area of 147,516 km² [24]. It has seven administrative provinces, within which lies 753 municipalities (6 metropolitan cities, 11 sub-metropolitan cities, 276 urban municipalities, 460 rural municipalities) [24]. Nepal has three ecological belts- Mountain, Hill, and Terai. Based on the 2021 Census, the total population of Nepal was 29,164,578 of which 14,911,027 (51.1 %) were females and 14,253,551 (48.9 %) were males [24]. Based on the Nepal Human Development Report 2020, Nepal was ranked 147th based on the human development index (HDI) with an overall HDI of 0.587 (rural: 0.647; urban: 0.561)[25]. According to the Global Hunger Index, Nepal has a ranking of 19.5, implying moderate food insecurity and undernourishment risk [26].

Sampling and data collection in NDHS

The NDHS uses a two-stage stratified cluster sampling design. In the first stage, wards were selected as primary sampling units using probability proportional to size. In the second stage, households within

each cluster were selected through systematic random sampling. Selected households were contacted through in-person visits, during which trained field enumerators listed all usual residents and visitors. All women aged 15–49 years who were either permanent residents or stayed in the selected household the night before the survey were invited to participate. In a sub-sample of households, men aged 15–49 years were also invited. In total, the 2022 NDHS interviewed 13,786 households out of 13833 occupied household (response rate: 99.7%). This yielded completed interviews from 14,845 women, 4,913 men, and proxy-reported information for 2,643 children under five [13]. Data collection was done in NDHS using standardized questionnaires (household, women's, men's, and biomarker questionnaires), administered through face-to-face interviews using tablet-based data entry. In this study, we analyzed the data of 2,395 children aged 6-59 months using the NDHS 2022 dataset.

Measures

Outcome variables

The primary outcome was “Co-existence of undernutrition (stunting, wasting or underweight) and anemia”. Anemia was assessed using altitude-adjusted hemoglobin levels and a child was considered anemic if the level was less than 11.0 g/dL [13]. A child is classified to have undernutrition if they exhibit either stunting, wasting, underweight, or any of these conditions, as based on the composite index of anthropometric failure (CIAF) concept [27]. In the NDHS survey, weight was measured using SECA scales (model SECA874U). Height and length were measured with a measuring board (Shorrboard®). Anemia was measured with blood specimens via finger-prick or heel-prick (for children 6-12 months of age). Hemoglobin was analyzed using a portable HemoCue® device [13].

Children whose height-for-age z-score is below minus two standard deviations (–2 SD) from the median of the internal reference population were defined as stunted [24]. Children whose weight-for-height z-score is below –2 SD from the median of the reference population were defined as wasted [28]. Children whose weight-for-age z-score is below –2 SD from the median of the reference population were defined as underweight [28].

The outcome variable co-existence of undernutrition and anemia consists of four categories a) Normal (having neither undernutrition nor anemia) b) only undernutrition (having only undernutrition) c) only anemia (having only anemia) and d) co-existence of undernutrition-anemia (having both undernutrition and anemia).

Exposure variables

The exposure variables for this study were wealth quintile (poorest/poorer/middle/richer/richest), maternal body mass index (underweight/normal/overweight or obese), mother's education (no

education/ basic level education/Secondary and higher level), mother's exposure to health programs in television and radio (yes/no), and mother's participation in household decision-making (yes/no). The definition of each exposure variable is explained in Table 1 below:

Table 1: List of exposure variables with their definition

Variables	Definition
Age of child	The age of the child was collected in months. We recategorized it into 6–12 months (late infant), 13–36 months (toddler) and 37–59 months (pre-school).
Wealth quintile	The wealth quintile measures the economic status of the household, and it is classified into poorest, poorer, middle, richer, and richest based on the wealth index. The wealth index included in the NDHS is a composite measure constructed using principal component analysis based on household assets, housing materials, water and sanitation facilities, and ownership of durable goods; households were assigned to five wealth quintiles (poorest to richest).
Mother's education	Mother's education was classified into basic education (grades 1 to 8), secondary level education (grades 9 to 12), or higher education (grades 13 and above) based on the DHS questionnaire.
Mother's exposure to Health program in TV/radio	The women were considered to have exposure to specific health programs if they heard or saw health programs on the radio or television (TV). This is measured based on the DHS questionnaire. These programs included eight different health-related programs broadcasted from radio and TV namely <i>Jana Swasthya Bahas</i> (Public Health Debate) television program, <i>Jeevan Chakra</i> (Life Cycle) television serial, <i>Jana Swasthya</i> (Public Health) radio program, <i>Swasthya Gatibidhi</i> (Health Affairs) radio program, <i>Eak Dui Tin Sunau eekai Chhin</i> (Listen for a While) radio program, <i>Bhanchhin Ama</i> (Mother Says) radio program, <i>Hello Bhanchin Ama</i> (Mother Says Hello) radio program and <i>Jeewan Rakshya</i> (Save Life COVID- response) radio programs [29].
Mother's participation in household decision-making	A composite variable measured from the mother's participation (alone or with her husband) in making three household decisions (access to healthcare, major household purchases, and visits her family or relatives) grouped into no participation or participation in decision-making [30,31].
Mother's body mass index (BMI)	The mother's BMI was classified as underweight, normal, overweight, and obese. The mother was considered underweight if BMI is less than 18.5 kg/m ² , normal if BMI is between 18.5 to 24.9 kg/m ² , overweight if her BMI is between 25 to 29.9 kg/m ² , and obese if her BMI is greater than 30.0 kg/m ² [32].

Potential Confounding variables

Based on the existing literature, some variables were found to be potential confounders in the association between outcome and exposure. Confounding variables included sociodemographic variables consisting of place of residence (rural/urban), and ecological belt (mountain/hill/terai), child-related variable consisting of sex of the child (male/female), and parents-related variables consisting of parity (Primipara/Multipara), mother's anemia status (a mother was considered anemic if the level was less than 11.0 g/dL. [13]) and father's education (no education/basic level/secondary and higher level).

Statistical analysis

We conducted pre-analytical processing and statistical analysis using R version 4.3.2 [33] and R studio [34]. We carried out weighted descriptive and inferential analysis using the “survey” package to address complex survey design and non-response. Parametric numerical variables were presented as mean (standard deviation) and non-parametric numerical variables as median (interquartile range). We presented categorical variables as frequency, percent, and their 95% confidence interval (CI). We computed the prevalence and 95% CI using the Wilson method.

We performed multivariable multinomial logistic regression to determine the association between the coexistence of undernutrition and anemia (none/undernutrition only/anemia only/coexistence) and the exposure variables listed in table 1, adjusting for potential confounding variables (place of residence, ecological belt, age of the child, sex of the child, mother's parity, father's education). We performed binary logistic regression to determine the association between undernutrition or anemia with the exposure variables. We checked for multicollinearity using variance inflation factor (VIF) and removed province (VIF= 13.31, greater than the cutoff of 10) from the regression models, as the information provided by the province was contained by the region and urban status variables. All regression analyses were conducted using complete cases after excluding observations with missing data. We calculated and presented crude odds ratio (cOR) and adjusted odds ratio (aOR) and their 95% CI.

Results

Table 2 presents the sociodemographic characteristics of 6-59 months of children stratified by age. Of the total U-5 children, 48.5% were female. Age distribution showed 10.4% were 6-12 months old, 45.8% were 13-36 months old, and 43.8% were 37-59 months old. Most children (59.1%) were from the Terai region, while the fewest (5.6%) were from the mountain region. The ratio of children from urban to rural areas was 2:1. By province, the majority were from Madhesh (26.4%), followed by Koshi (17.9%), Lumbini (16.7%), and Bagmati (15.4%).

Table 2: Characteristics of 6-59 months children

Characteristics	Overall, %	6-12 months children, %	13 to 36 months children, %	37-59 months children, %
Total unweighted frequency, <i>n</i>	2395	256	1092	1047
Total weighted frequency, <i>n</i>	2335	244	1069	1022
Age of children in months, %		10.4	45.8	43.8
Sex, %				
Male	51.5	52.6	51.4	51.3
Female	48.5	47.4	48.6	48.7
Ecological region, %				
Mountain	5.6	5.6	5.6	5.6
Hill	35.3	41.7	34.4	34.7
Terai	59.1	52.7	60	59.8
Type of place of residence, %				
Urban	64.3	63.1	64.2	64.7
Rural	35.7	36.9	35.8	35.3
Province, %				
Koshi	17.9	18.7	17.2	18.5
Madhesh	26.4	20.7	27.2	27
Bagmati	15.4	20	15.8	13.9
Gandaki	7.2	6.7	6.1	8.5
Lumbini	16.7	15.6	17.1	16.6
Karnali	7.7	9.8	7.1	7.8
Sudurpashchim	8.6	8.5	9.6	7.7
Wealth quintile, %				
Poorest	24.5	22.2	24.4	25
Poorer	22	19.9	23.5	20.9
Middle	20.7	19.6	19.6	22
Richer	18.1	19.1	19.2	16.8
Richest	14.7	19.2	13.3	15.2
Mother's education, %				
No education	22.2	15.3	23.1	22.9
Basic	35.9	32.5	35.7	37
Secondary and higher	41.9	52.2	41.2	40

<i>Missing, n</i>	55	0	14	40
Father's education, %				
No education	10.7	6.1	10.8	11.8
Basic	41	42.2	41.1	40.7
Secondary and higher	48.3	51.7	48.1	47.6
<i>Missing, n</i>	322	27	135	160
Mother's age at delivery, %				
<20 years	19.8	18.3	19.2	20.8
20-34 years	76.3	78.9	77.7	74.2
35-49 years	3.9	2.8	3.1	5
<i>Missing, n</i>	259	21	104	134
Parity, %				
Primipara	32.1	41.3	37.3	24.2
Multipara	67.9	58.7	62.7	75.8
<i>Missing, n</i>	259	21	104	134
Mother's BMI, %				
Underweight (<18.5 kg/m ²)	14.6	12.2	17.8	11.9
Normal (18.5 to 24.9 kg/m ²)	61.7	58.1	62.2	62.1
Overweight or obese ($\geq$ 25 kg/m ²)	23.7	29.7	20	26
<i>Missing, n</i>	399	22	185	192
Mother's participation in household decision-making, %				
No participation	33.5	32.5	34.6	32.6
Participation	66.5	67.5	65.4	67.4

n: weighted frequency; %: weighted percentage

Missing are presented as numbers

All percentages are column percentage

Figure 1 presents the prevalence of stunting, wasting and underweight among Nepalese children aged 6-59 months. The prevalence of stunting, wasting, and underweight were 25.9% (95% CI: 23.6 to 28.3), 7.8% (95% CI: 6.5 to 9.3), and 19.8% (95% CI: 17.6 to 22.1), respectively. Of the total children, 3.2% (95% CI: 2.5 to 4.3) had all three conditions, 13.5% (95% CI: 11.9 to 15.3) had two of the three conditions, and 16.8% (95% CI: 15.2 to 18.5) had one of the three conditions.

[Figure 1 here]

Figure 1: Prevalence of co-existence of different forms of undernutrition among A) children aged 6-12 months, B) children aged 13-36 months, C) children aged 37-59 months and D) all children aged 6-59 months.

The prevalence of anemia was 43.4% (95% CI: 40.9 to 45.9), which included 18.8% (95% CI: 16.8 to 20.9) with moderate to severe anemia and 24.6% (95% CI: 22.7 to 26.6) with mild anemia. The prevalence of moderate to severe anemia was highest in late infants, accounting for 39.9% (95% CI: 33.2 to 47.1) and decreased to 8.4% (95% CI: 6.7 to 10.8) in preschools (**Figure 2**).

[Figure 2 here]

Figure 2: Prevalence of anemia among A) children aged 6-12 months, B) children aged 13-36 months, C) children aged 37-59 months and D) all children aged 6 – 59 months.

The prevalence of co-existence of stunting-anemia, wasting-anemia, underweight-anemia, undernutrition-anemia were 12.5% (95% CI: 10.9 to 14.3), 4.0% (95% CI: 2.9 to 4.9), 9.7% (95% CI: 8.7 to 11.7), and 16.0% (95% CI: 14.2 to 18.0) respectively (**Figure 3A-D**). Of total children, 1.9% (95% CI: 1.3 to 2.6) had all three undernutrition conditions and anemia, 6.4% (95% CI: 5.3 to 7.8) had two undernutrition conditions and anemia, and 7.6% (95% CI: 6.6 to 8.9) had one undernutrition condition and anemia. (**not presented in tables**)

[Figure 3 here]

Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition

Table 3 presents the prevalence of co-existence of anemia and undernutrition by age group and sex of child. The prevalence of co-existence of undernutrition and anemia was highest in children 13-36 months (overall: 22.0%, male: 20.4%, female: 23.7%) followed by 6-12 months old children (overall: 16.2%, male: 17.4%, female: 14.9%). Prevalence of anemia only was highest in 6-12 months (overall: 54.9%, male: 50.2%, female: 60.1%) and only undernutrition in 37-59 months old children (overall: 23.5% male: 21.0%, female: 26.2%).

Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

Characteristic	6-59 months children, % (95%CI)	6-12 months children, % (95%CI)	13-36 months children, % (95%CI)	37-59 months children, % (95%CI)
All children				
None	39.1 (36.6 to 41.7)	25.0 (19.4 to 31.6)	33.6 (30.5 to 36.9)	48.2 (44.3 to 52.1)
Only Anemia	27.4 (25.3 to 29.6)	54.9 (48.3 to 61.3)	29.5 (26.4 to 32.7)	18.6 (15.8 to 21.8)
Only undernutrition	17.5 (15.6 to 19.6)	3.9 (1.99 to 7.49)	14.9 (12.5 to 17.6)	23.5 (20.6 to 26.8)
Co-existence of undernutrition and anemia	16.0 (14.1 to 18.0)	16.2 (11.7 to 22.0)	22.0 (19.1 to 25.3)	9.6 (7.89 to 11.7)
Male children				
None	39.5 (36.0 to 43.1)	29.5 (21.0 to 39.6)	33.5 (29.0 to 38.4)	48.2 (43.0 to 53.5)
Only Anemia	27.9 (25.1 to 30.9)	50.2 (41.1 to 59.2)	30.7 (26.6 to 35.1)	19.6 (15.9 to 23.9)
Only undernutrition	16.5 (14.2 to 19.2)	2.9 (0.94 to 8.83)	15.4 (12.3 to 19.2)	21.0 (17.2 to 25.5)
Co-existence of undernutrition and anemia	16.1 (13.7 to 18.8)	17.4 (11.3 to 25.9)	20.4 (16.9 to 24.5)	11.2 (8.71 to 14.2)
Female children				
None	38.7 (35.4 to 42.1)	20.1 (13.3 to 29.3)	33.8 (29.5 to 38.4)	48.1 (43.0 to 53.3)
Only Anemia	26.8 (23.8 to 30.1)	60.1 (49.2 to 70.0)	28.2 (23.9 to 33.1)	17.7 (13.8 to 22.3)
Only undernutrition	18.6 (15.9 to 21.5)	4.9 (2.09 to 11.2)	14.3 (11.2 to 18.1)	26.2 (21.7 to 31.2)
Co-existence of undernutrition and anemia	15.9 (13.4 to 18.7)	14.9 (9.01 to 23.5)	23.7(19.3 to 28.7)	8.0 (5.72 to 11.1)

%: weighted percent; CI: confidence interval; All percentages in the table are row percentage

Figure 4 and **Table 4** present the distribution of anemia, undernutrition and co-existence across categories of quintile wealth, mother's education, household decision-making and mother's exposure to health programs on TV and radio. The prevalence of undernutrition and co-existence was highest in poorest wealth quintile and lowest in richest wealth quintile, and highest in children whose mother had no education.

[Figure 4 here]

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables (vertical bars represent 95% CIs)

Table 4: Distribution of undernutrition, anemia and their co-existence across exposure variables

Characteristics	6-12 months children*			13-36 months children*			37-59 months children*		
	Anemia only, %	Undernutrition only, %	Co-existence, %	Anemia only, %	Undernutrition only, %	Co-existence, %	Anemia only, %	Undernutrition only, %	Co-existence, %
Wealth quintile									
Poorest	55.9	4.3	19.0	24.2	19.7	26.6	12.3	30.9	13.2
Poorer	51.7	1.8	15.2	30.3	16.4	25.4	17.6	27.4	11.0
Middle	60.9	3.3	12.2	31.0	12.3	21.3	24.4	27.3	10.1
Richer	56.9	4.0	24.0	35.2	12.5	18.5	29.5	13.0	5.9
Richest	48.8	6.2	10.4	27.2	10.4	13.7	10.2	12.4	5.3
Mother's education									
No education	51.7	8.1	20.9	27.5	15.1	35.5	21.9	29.0	15.6
Basic	51.1	1.1	22.4	27.7	16.1	22.9	17.7	27.9	9.4
Secondary and higher	58.1	4.4	11.0	32.3	13.8	13.9	18.0	15.9	7.2
Participation in household decision-making	52.9	4.7	14.9	28.5	14.9	21.3	17.9	22.2	9.4
Exposure to health programs on TV/radio	64.3	4.7	11.9	24.6	15.5	14.2	16.6	26.7	10.0
Mother's BMI									
Underweight	50.3	6.9	25.3	22.1	22.5	33.4	28.3	22.5	13.2
Normal	58.0	2.9	17.1	30.3	14.4	21.1	15.7	26.2	10.8
Overweight or obese	46.1	5.0	13.9	37.7	10.2	12.9	22.1	18.8	5.5

#: weighted percentage

* Row percentage

Table 5 presents the factors associated with undernutrition and anemia among children aged 6-59 months. In univariable regression, undernutrition was significantly associated with wealth quintile, mother's education, age of child, and maternal nutritional status. In multivariable regression, undernutrition was found to be associated with age of child, wealth quintile, mother's education, and maternal nutrition status after adjusting for confounding variables. The odds of having undernutrition

were 1.96 times among 13-36 months children and 1.70 times in the 37-59 months children compared to 6-12 months children. The odds of undernutrition were 44% (aOR: 0.56, 95%CI: 0.34 to 0.92) lower in children from the richer wealth quintile compared to the poorest, 36% (aOR: 0.64, 95%CI: 0.45 to 0.93) lower in the child whose mother had secondary or higher-level education. We did not find a significant association between undernutrition and mother participation in household decision-making and exposure to health programs on TV/radio.

Table 5: Factors associated with undernutrition among children aged 6-59 months

Characteristic	Presence of undernutrition, n (%) *	Unadjusted models			Adjusted model [#]		
		cOR	95% CI	p-value	aOR	95% CI	p-value
Age of child							
6-12 months	49 (20.1)	Ref			Ref		
13-36 months	394 (36.9)	2.32	1.67 to 3.22	<0.001	1.96	1.33 to 2.90	<0.001
37-59 months	339 (33.2)	1.97	1.39 to 2.81	<0.001	1.70	1.12 to 2.59	0.013
Wealth quintile							
Poorest	246 (43.1)	Ref			Ref		
Poorer	195 (38.1)	0.81	0.62 to 1.06	0.122	0.93	0.67 to 1.30	0.680
Middle	162 (33.6)	0.67	0.50 to 0.88	0.005	0.78	0.54 to 1.13	0.194
Richer	109 (25.7)	0.46	0.32 to 0.66	<0.001	0.53	0.34 to 0.82	0.005
Richest	70 (20.2)	0.33	0.22 to 0.50	<0.001	0.56	0.34 to 0.92	0.024
Mother's education							
No education	234 (46.3)	Ref			Ref		
Basic	301 (36.7)	0.67	0.52 to 0.88	0.003	0.87	0.65 to 1.16	0.332
Secondary and higher	231 (24.2)	0.37	0.28 to 0.49	<0.001	0.64	0.45 to 0.93	0.019
Mother's Participation in household decision making							
No participation	279 (35.7)	Ref			Ref		
Participation	503 (32.4)	0.86	0.70 to 1.07	0.173	0.82	0.63 to 1.06	0.133
Exposure to health programs on TV/radio							
No	615 (34.5)	Ref			Ref		
Yes	145 (31.0)	0.85	0.67 to 1.09	0.199	0.92	0.69 to 1.24	0.595
Mother's BMI							
Normal	411 (34.5)						
Underweight	132 (46.6)	1.66	1.23 to 2.23	<0.001	1.48	1.09 to 2.01	0.013
Overweight or obese	106 (23.1)	0.57	0.43 to 0.75	<0.001	0.75	0.56 to 1.01	0.057

n: weighted frequency; %: weighted percent; Ref: reference group; cOR: crude odds ratio; aOR: adjusted odds ratio; CI: confidence interval; * Row percent
[#] adjusted for sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

Table 6 presents the factors associated with anemia among children aged 6-59 months. The factors associated were age of the child, wealth quintile, and anemia status of the mother. The odds of having anemia was 41% (aOR: 0.59, 95%CI: 0.36 to 0.97) lower among children belonging to richest quintile compared to poorest. The odds of having anemia was 64% (aOR: 0.36, 95%CI: 0.24 to 0.53) lower among children aged 13-36 months and 87% (aOR: 0.13; 95%CI: 0.09 to 0.20) lower child aged 37-59 months compared to child aged 6-12 months after adjusting for potential confounding variables.

291 **Table 6: Factors associated with anemia among children aged 6-59 months**

Characteristic	Prevalence of anemia, n (%) *	Unadjusted models			Adjusted model [#]		
		cOR	95% CI	p-value	aOR	95% CI	p-value
Age of child							
6-12 months	173 (71.1)	Ref			Ref		
13-36 months	551 (51.5)	0.43	0.31 to 0.60	<0.001	0.36	0.24 to 0.53	<0.001
37-59 months	289 (28.3)	0.16	0.11 to 0.23	<0.001	0.13	0.09 to 0.20	<0.001
Wealth quintile							
Poorest	239 (41.8)	Ref			Ref		
Poorer	233 (45.5)	1.16	0.88 to 1.53	0.291	0.88	0.63 to 1.23	0.451
Middle	222 (46.0)	1.19	0.90 to 1.56	0.215	0.89	0.61 to 1.30	0.531
Richer	209 (49.2)	1.35	1.02 to 1.80	0.038	1.07	0.72 to 1.60	0.725
Richest	110 (31.9)	0.65	0.46 to 0.93	0.02	0.59	0.36 to 0.97	0.037
Mother's education							
No education	265 (52.4)	Ref			Ref		
Basic	347 (42.4)	0.67	0.52 to 0.86	0.002	0.80	0.57 to 1.13	0.210
Secondary and Higher	388 (40.6)	0.62	0.48 to 0.80	<0.001	0.84	0.56 to 1.26	0.409
Mother's Participation in decision making							
No participation	365 (46.6)	Ref			Ref		
Participation	648 (41.7)	0.82	0.65 to 1.03	0.095	0.76	0.55 to 1.05	0.099
Exposure to health programs on TV/radio							
No	813 (45.6)	Ref			Ref		
Yes	179 (38.4)	0.74	0.59 to 0.94	0.012	1.00	0.74 to 1.35	0.990
Mother's BMI							
Underweight	149 (52.5)	1.46	1.07 to 1.98	0.016	1.20	0.84 to 1.71	0.319
Normal	516 (43.2)	Ref			Ref		
Overweight or obese	189 (41.1)	0.92	0.70 to 1.21	0.548	1.19	0.87 to 1.63	0.286

n: weighted frequency; %: weighted percent

* Row percent

Ref: reference group; cOR: crude odds ratio; aOR: adjusted odds ratio; CI: confidence interval

adjusted for sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, mother's anemia status, and adjusted for each other

Table 7 presents the factors associated with co-existence of undernutrition and anemia among children aged 6-59 months. The co-existence of undernutrition and anemia was significantly associated with the richest wealth quintile, the mother's education status, the mother's BMI, and presence of anemia in children. The children from the richest wealth quintile were 53% (aOR: 0.47; 95%CI: 0.26 to 0.86) less likely to have co-existence of undernutrition and anemia than children from the poorest wealth quintile. The children whose mothers had secondary or higher-level education had 48% (aOR: 0.52; 95%CI: 0.32 to 0.86) lower odds of having co-existence compared to children whose mothers had no formal education. The odds of having co-existence of undernutrition and anemia was 34% (aOR: 0.66; 95%CI: 0.47 to 0.94) lower in the children whose mother participate in household decision-making after adjusting for confounding variables. The child whose mother was underweight was 1.80 times (95%CI:

1.20 to 2.70) more likely and mother with anemia were 2.43 times more likely to have co-existence compared to a child whose mother had normal nutrition status.

Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged 6-59 months

Exposure variables	Anemia only vs Normal			Undernutrition only vs Normal			Co-existence vs Normal		
	aOR [#]	95% CI	p-value	aOR [#]	95% CI	P-value	aOR [#]	95% CI	p-value
Age of child (ref: 6-12 months child)									
13-36 months	0.37	0.25 to 0.54	<0.001	3.02	1.32 to 6.91	0.009	0.68	0.42 to 1.11	0.123
37-59 months	0.16	0.11 to 0.24	<0.001	3.16	1.39 to 7.20	0.006	0.21	0.12 to 0.35	<0.001
Wealth Quintile (ref: Poorest)									
Poorer	0.94	0.63 to 1.40	0.77	0.98	0.65 to 1.48	0.928	0.8	0.51 to 1.25	0.325
Middle	0.98	0.64 to 1.49	0.917	0.83	0.53 to 1.30	0.412	0.61	0.37 to 1.01	0.054
Richer	1.04	0.67 to 1.61	0.869	0.48	0.28 to 0.81	0.006	0.58	0.34 to 0.99	0.048
Richest	0.49	0.30 to 0.79	0.004	0.38	0.21 to 0.67	0.001	0.47	0.26 to 0.86	0.015
Mother's education (ref: no education)									
Basic	0.87	0.60 to 1.27	0.473	1.00	0.67 to 1.49	0.996	0.70	0.46 to 1.06	0.094
Secondary and higher	1.02	0.67 to 1.56	0.923	0.87	0.54 to 1.38	0.543	0.52	0.32 to 0.86	0.010
Participation in HH decision-making (ref: nonparticipation)	0.69	0.51 to 0.93	0.016	0.73	0.52 to 1.02	0.064	0.66	0.47 to 0.94	0.02
Exposure to health programs on TV/radio (ref: No)	1.04	0.77 to 1.42	0.783	1.00	0.70 to 1.42	0.996	0.89	0.59 to 1.34	0.572
Mother's BMI (ref: normal weight)									
Underweight	1.17	0.80 to 1.71	0.426	1.44	0.96 to 2.16	0.075	1.80	1.20 to 2.70	0.005
Overweight and obese	1.30	0.97 to 1.74	0.076	0.89	0.63 to 1.28	0.539	0.88	0.59 to 1.32	0.529

Normal: participants having neither undernutrition nor anemia

ref: reference group; **aOR:** adjusted odds ratio; **CI:** confidence interval

Bold represents significance at 95% confidence level

[#] adjusted for sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, mother's anemia status, and adjusted for each other

Discussion:

Our analysis of a national survey revealed significant issues related to undernutrition, anemia and their co-existence among Nepalese children in 2022. Overall, the prevalence of undernutrition and anemia was 33.5% and 43.4%, respectively. The prevalence of co-existence between undernutrition and anemia was 16.0%. Specifically, the co-existence of stunting, wasting, and underweight with anemia was 12.5%, 4.0%, and 9.7%, respectively. Undernutrition demonstrated significant associations with age of child, household wealth, mother's education, and mother's BMI. Anemia was associated with both household wealth and maternal anemia status. The co-existence of undernutrition and anemia exhibited significant associations with wealth, maternal education, maternal involvement in household decision-making, parity, mother's

BMI, and the child's anemia status. However, none of the three outcomes (undernutrition, anemia, and their co-occurrence) were found to be associated with maternal exposure to health programs on television and radio.

A study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and among children aged 6-59 months (57.3%) [35]. These figures are higher than the prevalence of undernutrition among 6-59 months children in Nepal (33.5%) found in our study. Our findings indicate that Nepal's burden, though lower, remains substantial. These differences may reflect variations in economic development, food security, and maternal and child health interventions across countries [36].

The prevalence of anemia among 6-59 months children in our study (43.5%) was higher than the global prevalence (39.8%) but lower than that reported among African children (60.2%) [12]. Our study also demonstrates a 9.1 percentage point reduction in anemia prevalence between 2016 (52.6%) [37] and 2022, indicating progress in addressing this issue. Comparable rates have been observed in neighboring countries. A meta-analysis conducted by Kundu et al in Bangladesh reported a prevalence of 46.8% [38], which was similar to our finding. The latest National Family Health Survey (2019-21) in India reported a prevalence of 67.1%, which is higher than that observed in Nepal [39]. The decline in anemia prevalence in Nepal is encouraging but underscores the need for sustained efforts to address this condition, particularly among vulnerable populations [23].

A systematic review reported the pooled prevalence rates of wasting-anemia and stunting-anemia in least-developed countries to be 5.4% and 19.5%, respectively [40]. These rates are higher than our findings from Nepal, which were 4.0% and 12.5%, respectively.

The high prevalence of co-existence of undernutrition and anemia in our study (16%) suggests that affected children are caught in an interlinked cycle of micronutrient deficiencies. Undernutrition impairs iron absorption and utilization, while anemia exacerbates undernutrition by reducing the body's ability to effectively utilize nutrients.

The children from the richest wealth quintile had lower odds of having the co-existence of anemia and undernutrition compared to those from the poorest quintile. This wealth-related equity gap in undernutrition among children aged 6–59 months is attributed to household economic status, which influences food insecurity and health care utilization when a child is ill [41–43].

Children from families in which mothers participated in household decision-making and mothers with secondary or higher education were less likely to experience the co-existence of anemia and undernutrition, as well as anemia alone, compared to children without these conditions. The health and nutrition of the child largely depend on the mother, who is typically the primary caretaker and responsible for awareness of the child’s health. Analyses of DHS data from Nepal and India suggest that women’s decision-making authority (maternal autonomy in household decision-making) improves children’s nutritional status and reduces child mortality, even after controlling for education and wealth [44].

Strengths and Limitations

This study has several strengths. First, it utilized complex survey analysis to account for survey design and non-response rates. Second, the findings are generalizable to children aged 6-59 months in Nepal and to children of the same age group in demographically and economically similar countries, as the study used a nationally representative dataset.

This study has several limitations. First, due to the missing data, we could not include some of the important dietary factors like minimum dietary diversity score and child’s birth weight. Second, because of the cross-sectional nature of the study, the directionality of associations cannot be established. Third, the results from this study may not be generalizable to populations outside Nepal with different demographic and economic status. Lastly, the wide confidence intervals observed in the associations between children’s nutritional status and anemia may suggest limited statistical power, likely due to small sample sizes within certain subgroups.

Conclusion

The prevalence of undernutrition, anemia, and their co-existence remains high among children aged 6-59 months in Nepal. The co-existence of anemia and undernutrition was less likely among children from the highest wealth quintile, those whose mothers attained at least secondary level education, and those whose mothers participated in household decision-making. In contrast, co-existence was more likely among children whose mothers had poor maternal nutritional status. It is essential to implement

385 holistic programs and targeted interventions to empower poor families, improve mother's education,
386 strengthen women's roles in household decision-making, and ultimately improve the children's health.

388 **Data and analysis code availability**

389 We used publicly available data from Nepal Demographic and health survey database upon request
390 from the official website of "The DHS program". The dataset is available at:

391 https://dhsprogram.com/data/dataset/Nepal_Standard-DHS_2022.cfm?flag=1

392 The analysis code and data used for this analysis is available on GitHub:

393 <https://github.com/Adrisyaone/Undernutritiona-and-anemia>

396 **Conflict of interest**

397 The authors declare no conflicts of interest.

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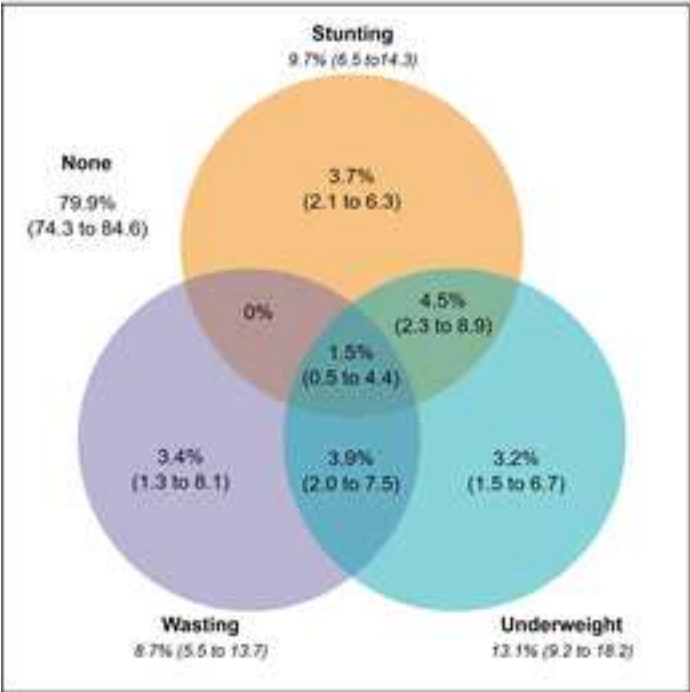
Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables

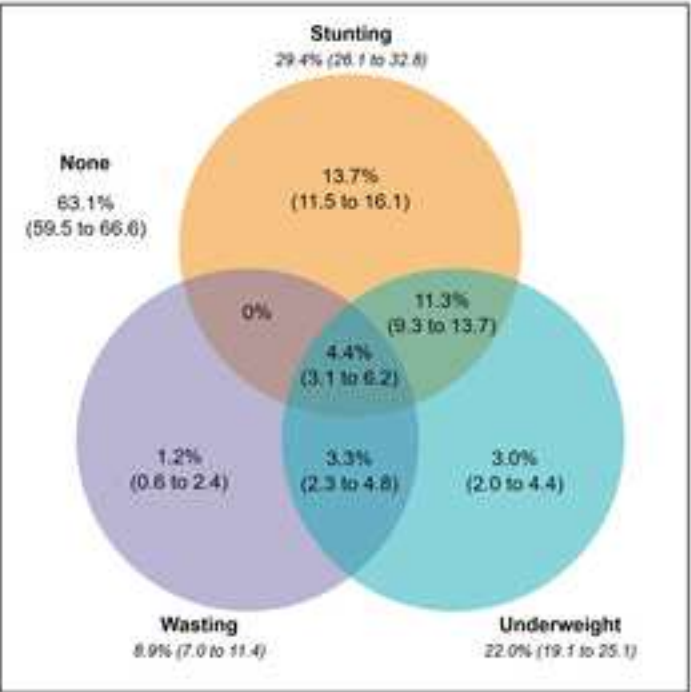
Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged 6-12 months, B) children

[Click here to access/download;Figure;Figure 1.tif](#)

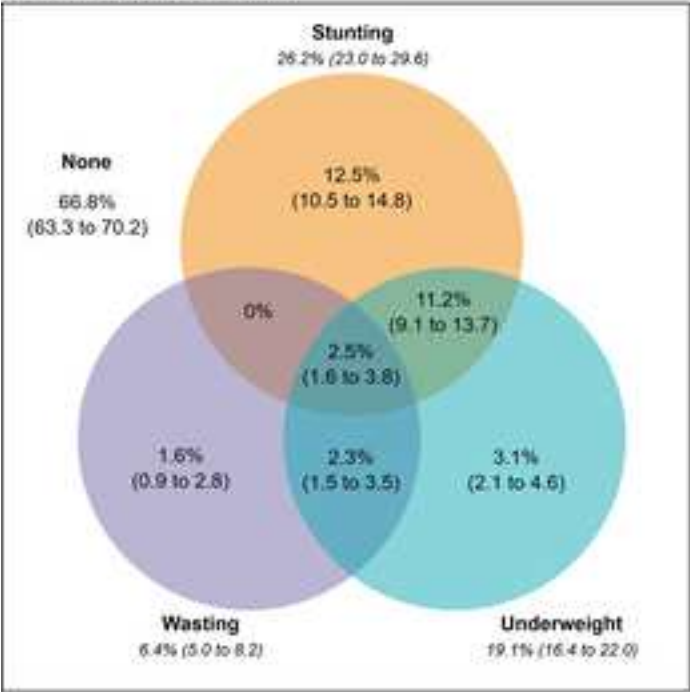
A. Children aged 6-12 months



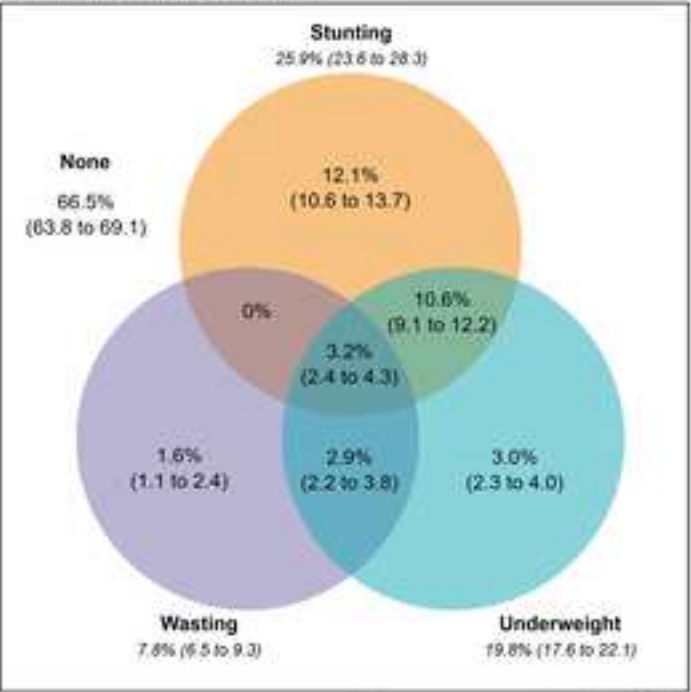
B. Children aged 13-36 months



C. Children aged 37-59 months

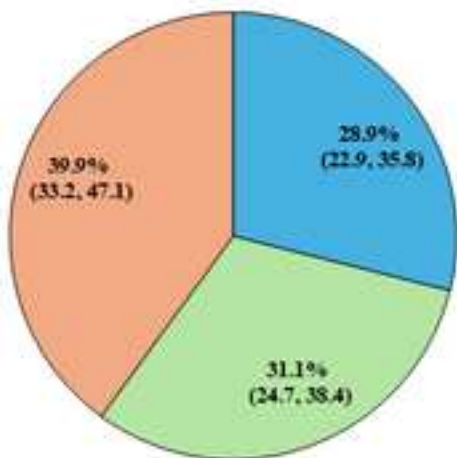


D. All children aged 6-59 months

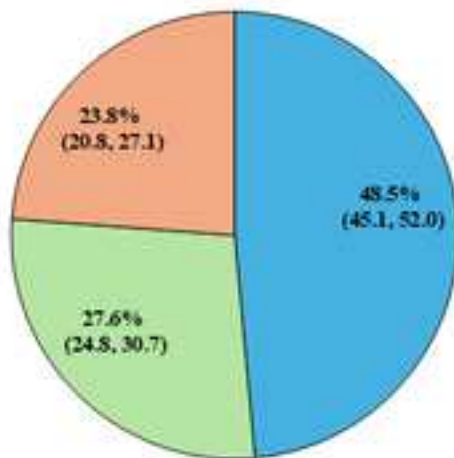


Values are expressed as % (95% CI)

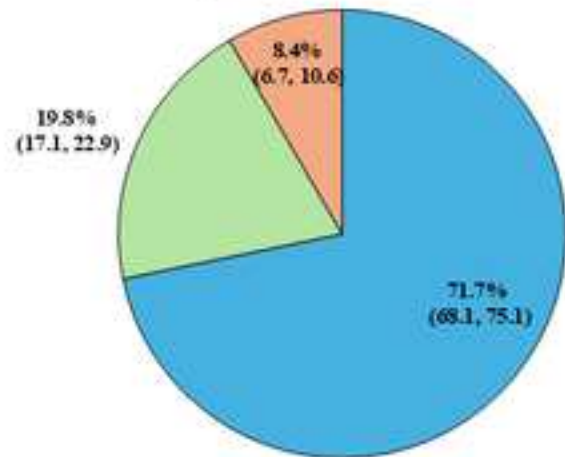
B. Children aged 6 months to 12 months



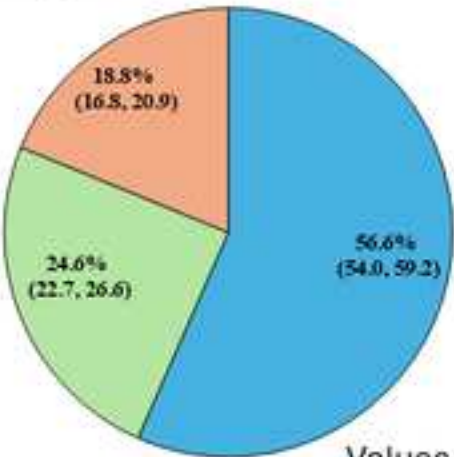
C. Children aged 13-36 months



D. Children aged 37-59 months



A. Children aged 6-59 months

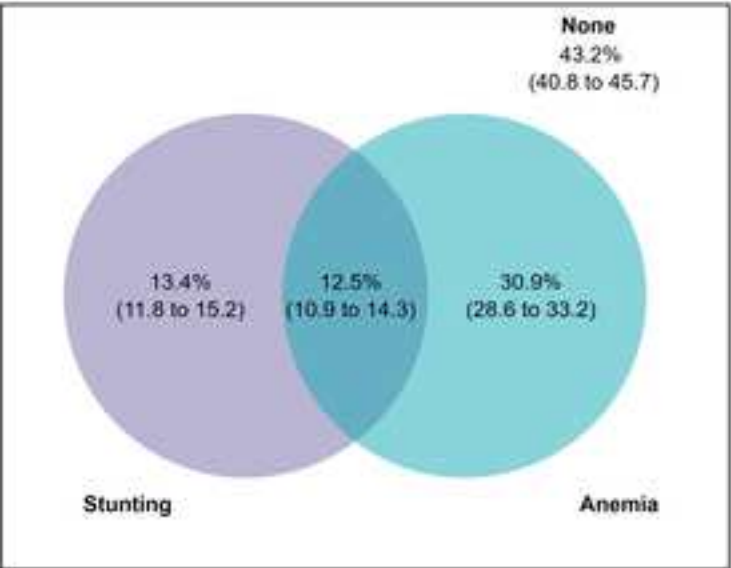


■ Normal
■ Mild
■ Moderate to severe

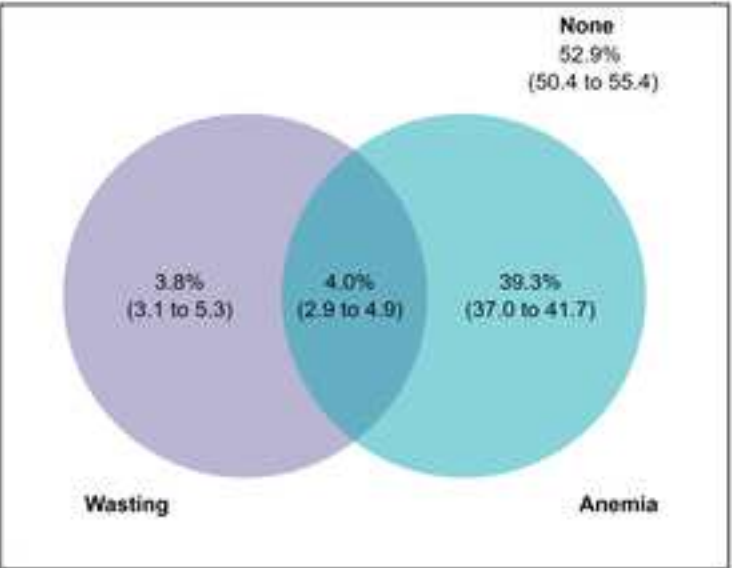
Values are expressed as % (95% CI)

Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition

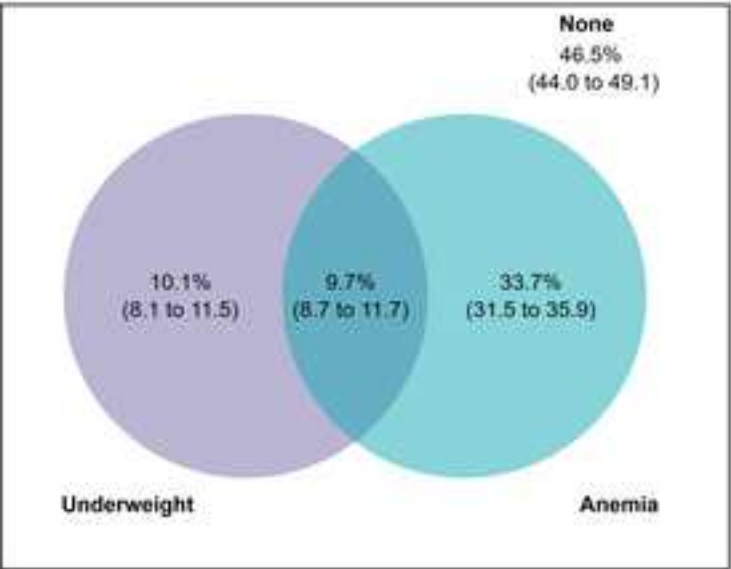
A. Co-existence of stunting and anemia



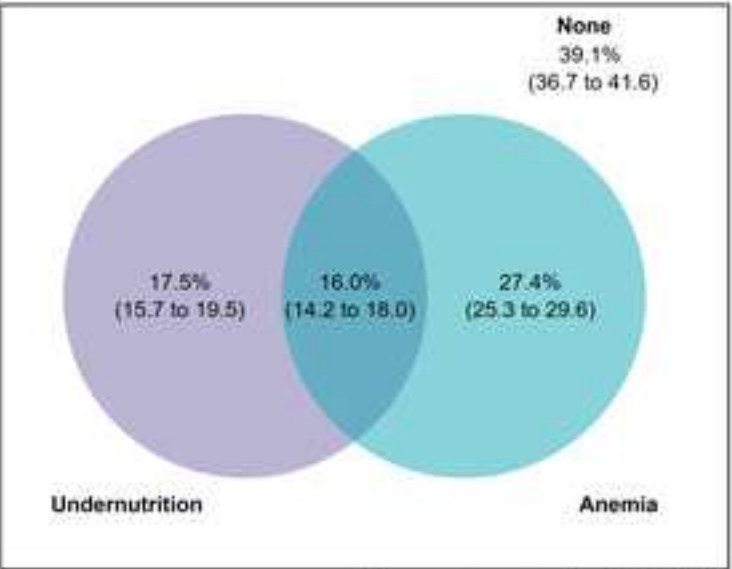
B. Co-existence of wasting and anemia



C. Co-existence of underweight and anemia



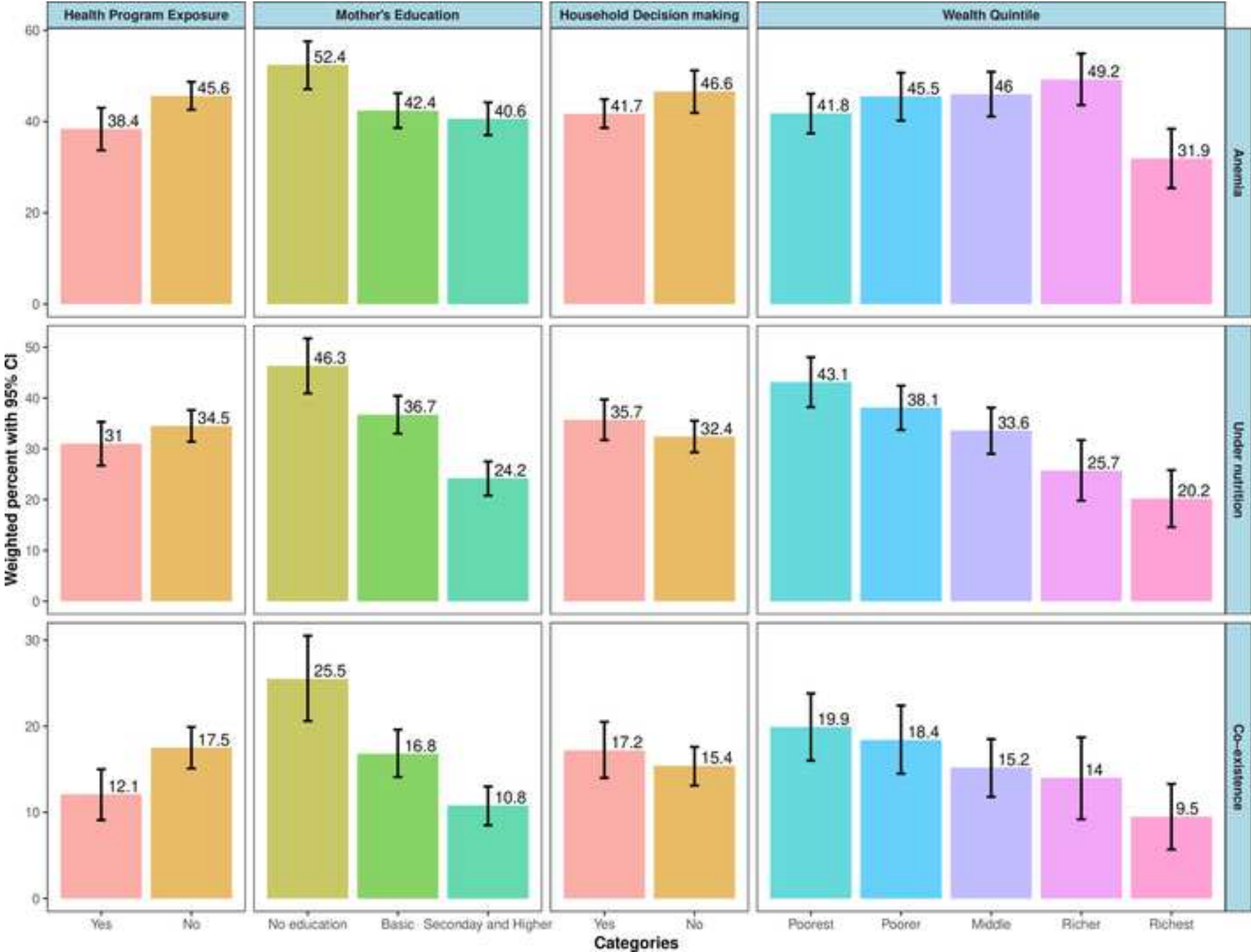
D. Co-existence of undernutrition and anemia



Values are expressed as % (95% CI)

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables

[Click here to access/download;Figure;Figure 4.tif](#)



1 **Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59**
2 **Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey**

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Abstract

Introduction: Undernutrition and anemia among children aged 6 to 59 months are significant public health issues in developing countries like Nepal. The co-existence of these conditions impacts childhood development. This study aimed to determine the prevalence of undernutrition and anemia, assess their co-existence, and identify contributing factors among children aged 6-59 months in Nepal.

Methods: We conducted a secondary analysis using data from 2022 Nepal Demographic and Health Survey (NDHS 2022), a nationally representative cross-sectional survey. The outcome variables were undernutrition, anemia, and their co-existence. ~~Undernutrition was defined as the Children aged 6-59 months were considered undernutrition if they exhibited presence of either~~ stunting, wasting, underweight or any combination of these conditions. Anemia was defined as hemoglobin levels <11.0 g/dL (adjusted for altitude). We applied multivariable multinomial logistic regressions to determine factors associated with co-existence, and multivariable logistic regression to assess factors associated with undernutrition and anemia separately. We presented the results from the regression analysis using adjusted odds-ratio (aOR) and 95% confidence intervals (CI).

Results: Among 2,395 children, the weighted prevalence of undernutrition was 33.5%, anemia was 43.4%, and co-existence of undernutrition and anemia was 16.0%. Children from the richest wealth quintile, whose mothers had at least secondary education, and those whose mother participated in household decision-making had 53% (aOR: 0.47; 95% CI: 0.26 to 0.86), 48% (aOR: 0.52; 95% CI: 0.32 to 0.8693), and 34% (aOR: 0.66; 95% CI: 0.47 to 0.94) lower odds of experiencing the co-existence of undernutrition and anemia compared to their counterparts. Children with underweight mothers had 80% (aOR: 1.80; 95% CI: 1.20 to 2.70) higher odds of co-existence compared to their counterparts.

Conclusion: The prevalence of undernutrition, anemia, and their co-existence among children was high in Nepal. Interventions that improve mother's education, strengthen their roles in the family, and enhance the household financial condition ~~of household~~ are essential to reduce these conditions and improve children's nutritional status.

Keywords: undernutrition; anemia; co-existence; 6-59-months children; Nepal; NDHS 2022

49 Introduction

50 Undernutrition including stunting, wasting, and being underweight is a major public health challenge
51 among children under 5 years (U-5) of age globally. This issue is particularly pronounced in developing
52 countries like Nepal [1–4]. There are three different causes of undernutrition – immediate, underlying,
53 and basic: The immediate causes include inadequate diet intake and repeated illnesses; The underlying
54 causes include suboptimal child feeding and caregiver practices from mothers, lack of access to
55 healthcare facilities, poverty, and food insecurity; ~~The~~ basic causes include limited information, political
56 and economic insecurity, gender inequality, and the occurrence of natural disasters [5].

57 Different ~~indices~~ of undernutrition measure different underlying development status. Stunting
58 indicates chronic malnutrition reflecting long-term nutrient deficiencies and is linked with delayed ~~and~~
59 ~~impaired~~ motor ~~development~~ and ~~impaired~~ cognitive development [6]. Wasting indicates acute
60 malnutrition associated with recent severe food shortages or illnesses leading to weight loss and is a
61 strong predictor of mortality [6]. Underweight combines information about linear growth obstruction
62 and weight for length or height [7].

63 Anemia, particularly iron deficiency anemia, is another common and pressing public health concern
64 among children U-5 in low-, middle- and high-income countries. It results from poor nutrition and has
65 severe adverse health consequences including impaired cognitive development, impaired immunity,
66 disability, and ~~an~~ increased risk of morbidity and mortality [8–10].

67 In 2022, the global prevalence of stunting among children under five years of age was 22.3%, while in
68 South Asia, it reached 31.8%. Additionally, the global prevalence of wasting was at 6.8%, with South
69 Asia reporting a prevalence of 14.8% [11]. In 2019, global anemia prevalence was 39.8% [12]. In Nepal,
70 the prevalence of stunting, and wasting among children under 5 was 25% and 8% respectively, and the
71 prevalence of anemia was 43% among children 6-59 months old [13].

72 The co-existence of undernutrition and anemia increases the risk of childhood morbidity and mortality
73 [14–16]. Almost fifty percent of deaths among children under the age of five are associated with
74 undernutrition, which predominantly occurs in low- and middle-income countries (LMICs). Notably,
75 88% of these countries (124 out of 141) experience multiple forms of malnutrition [17]. The
76 developmental, economic, social, and medical impacts of undernutrition and anemia among children
77 are serious and perpetuating, for individuals and their families, communities, and countries [7].

78 Furthermore, anemia and undernutrition are both concentrated in socioeconomically disadvantaged
79 groups, which share numerous multifaceted causes and complex interactions between diet,
80 transmissible illnesses, inadequate care and unhealthy household environments that adversely affect the
81 cognitive development and physical well-being of children [18–20]. For instance, lower

82 ~~mother's~~maternal education levels and socioeconomic status of households are linked with higher rates
83 of malnutrition among children [18].

84 Though the concerns on nutrition should be explored beyond any single form of malnutrition to the co-
85 existence of multiple forms, most studies in Nepal have focused on undernutrition and anemia
86 separately. A few studies that have examined both conditions together primarily reported that children
87 from poorer households, mothers with low education levels, and those with inadequate dietary diversity
88 were more likely to experience concurrent undernutrition and anemia [21,22]. Even when both
89 malnutrition and anemia were considered, anemia was often treated merely as a correlating factor or
90 predictor rather than as part of a combined outcome reflecting their simultaneous occurrences.
91 Moreover, there remains a gap in understanding how maternal factors (such as nutritional status and
92 exposure to health information) and household-level socioeconomic characteristics interact to influence
93 the co-existence of undernutrition and anemia among children under five in Nepal.

94 Therefore, this study aims to ascertain the prevalence of stunting, wasting, and underweight, as well as
95 their co-occurrence, and the co-existence of undernutrition and anemia in Nepal. It also assesses the
96 association of undernutrition, anemia, and co-existence of undernutrition with household economic
97 status, maternal education and nutritional status, and exposure to televised or broadcasted health
98 programs among children aged 6-59 months in Nepal. By identifying these factors, this study helps
99 inform the development of targeted interventions and policies designed to mitigate malnutrition within
100 this vulnerable population, thereby enhancing child health outcomes and alleviating the burden of
101 malnutrition in Nepal.

102 **Methods**

103 *Data source*

104 In this study, we analyzed data from the Nepal Demographic and Health Survey (NDHS) conducted in
105 2022 [23]. NDHS 2022 is the nationally representative survey implemented by New ERA under the
106 aegis of the Ministry of Health and Population (MoHP) with the technical support of ICF International
107 and funding from the United States Agency for International Development (USAID) [13]. The NDHS
108 follows standardized demographic and health survey (DHS) protocols for sampling, data collection, and
109 quality assurance.

110

111 *Ethical approval*

112 We received permission from the official website of “the DHS program”
113 (<https://www.dhsprogram.com>) to download and use (submitted on: 07/25/2023 and approved on
114 07/26/2023) NDHS 2022 dataset [23]. NDHS 2022 obtained ethical approval from the Institutional
115 Review Board of ICF International, United States of America (Reference number:
116 180657.0.001.NP.DHS.01, Date: 28th April 2022). and the ethical review board of Nepal Health
117 Research Council (Reference number: 678, Date: 30th September 2021) [13]. In the NDHS 2022,
118 informed consent was taken from the participants before enrolling them into the study.

119

120 *Study setting*

121 This study used nationally representative data from Nepal, a landlocked country located in Southeast
122 Asia with an area of 147,516 km² [24]. It has seven administrative provinces, within which lies 753
123 municipalities (6 metropolitan cities, 11 sub-metropolitan cities, 276 urban municipalities, 460 rural
124 municipalities) [24]. Nepal has three ecological belts- Mountain, Hill, and Terai. Based on the 2021
125 Census, the total population of Nepal was 29,164,578 of which 14,911,027 (51.1 %) were females and
126 14,253,551 (48.9 %) were males [24]. Based on the Nepal Human Development Report 2020, Nepal
127 was ranked 147th based on the human development index (HDI) with an overall HDI of 0.587 (rural:
128 ~~and rural and urban parts of Nepal of 0.647; and urban: 0.561-)respectively~~ [25]. According to the
129 Global Hunger Index, Nepal has a ranking of 19.5, implying moderate food insecurity and
130 undernourishment risk [26].

131

132 *Sampling and data collection in NDHS*

The NDHS uses a two-stage stratified cluster sampling design. In the first stage, wards were selected as primary sampling units using probability proportional to size. In the second stage, households within each cluster were selected through systematic random sampling. Selected households were contacted through in-person visits, during which trained field enumerators listed all usual residents and visitors. All women aged 15–49 years who were either permanent residents or stayed in the selected household the night before the survey were invited to participate. In a sub-sample of households, men aged 15–49 years were also invited. In total, the 2022 NDHS interviewed 13,786 households out of 13833 occupied household (response rate: 99.7%). This yielded ~~yielding~~ completed interviews from 14,845 women, 4,913 men, and proxy-reported information for 2,643 children under five [13]. Data collection was done in NDHS using standardized questionnaires (~~Household~~household, ~~Woman's~~women's, ~~Man's~~men's, and ~~Biomarker~~biomarker questionnaires), administered ~~via~~through face-to-face interviews using tablet-based data entry. In this study, we ~~will~~analyzed the data of 2,395 children aged ~~6-months to 59 months-years~~ using the NDHS 2022 dataset.

Measures

Outcome variables

The primary outcome was “Co-existence of undernutrition (stunting, wasting or underweight) and anemia”. Anemia was assessed using altitude-adjusted hemoglobin levels and a child was considered anemic if the level was less than 11.0 ~~gm/deciliter~~dL [13]. A child is classified to have undernutrition if they exhibit either stunting, wasting, underweight, or any of these conditions, as based on the composite index of anthropometric failure (CIAF) concept [27]. In the NDHS survey, weight was measured using SECA scales (model SECA874U). Height and length were measured with a measuring board (Shorrboard â). Anemia was measured with blood specimens via finger-prick or heel-prick (for children 6-12 months of age). Hemoglobin was analyzed using a portable HemoCueâ device [13]. Children whose height-for-age z-score is below minus two standard deviations (–2 SD) from the median of the internal reference population were defined as stunted [24]. Children whose weight-for-height z-score is below –2 SD from the median of the reference population were defined as wasted [28]. Children whose weight-for-age z-score is below –2 SD from the median of the reference population were defined as underweight [28].

The outcome variable co-existence of undernutrition and anemia consists of four categories a) Normal (having neither undernutrition nor anemia) b) only undernutrition (having only undernutrition) c) only anemia (having only anemia) and d) co-existence of undernutrition-anemia (having both undernutrition and anemia).

Exposure variables

The exposure variables for this study were wealth quintile (poorest/poorer/middle/richer/richest), maternal body mass index (underweight/normal/overweight or obese), mother's education (no education/ basic level education/Secondary and higher level), mother's exposure to health programs in television and radio (yes/no), and mother's participation in household decision-making (yes/no). The definition of each exposure variable is explained in Table 1 below:

Table 1: List of exposure variables with their definition

Variables	Definition
Age of child	The <u>age</u> of the child was collected in months. We recategorized it into 6–12 months (late infant), 13–36 months (toddler) and 37–59 months (pre-school).
Wealth quintile	The wealth quintile measures the economic status of the household, and it is classified into poorest, poorer, middle, richer, and richest based on the wealth index. The wealth index included in the NDHS is a composite measure constructed using principal component analysis based on household assets, housing materials, water and sanitation facilities, and ownership of durable goods; households were assigned to five wealth quintiles (poorest to richest).
Mother's education	Mother's education was classified into basic education (grades 1 to 8), secondary level education (grades 9 to 12), or higher education (grades 13 and above) based on the DHS questionnaire.
Mother's exposure to Health program in TV/radio	The women were considered to have exposure to specific health programs if they heard or saw health programs on the radio or television (TV). This is measured based on the DHS questionnaire. These programs included eight different health-related programs broadcasted from radio and TV namely <i>Jana Swasthya Bahas</i> (Public Health Debate) television program, <i>Jeevan Chakra</i> (Life Cycle) television serial, <i>Jana Swasthya</i> (Public Health) radio program, <i>Swasthya Gatibidhi</i> (Health Affairs) radio program, <i>Eak Dui Tin Sunau eekai Chhin</i> (Listen for a While) radio program, <i>Bhanchhin Ama</i> (Mother Says) radio program, <i>Hello Bhanchin Ama</i> (Mother Says Hello) radio program and <i>Jeewan Rakshya</i> (Save Life COVID- response) radio programs [29].
Mother's participation in household decision-making	A composite variable measured from the mother's participation (alone or with her husband) in making three household decisions (access to healthcare, major household purchases, and visits her family or relatives) grouped into no participation or participation in decision-making [30,31].
Mother's body mass index (BMI)	The mother's BMI was classified as underweight, normal, overweight, and obese based on the body mass index of the mother . The mother will be <u>was</u> considered underweight if BMI is less than 18.5 kg/m ² , normal if BMI is between 18.5 to 24.9 kg/m ² , overweight if her BMI is between 25 to <u>-</u> 29.9 kg/m ² , and <u>-</u> and obese if her BMI is greater than 30.0 kg/m ² [32].

Potential Confounding variables

Based on the existing literature, some variables were found to be potential confounders in the association between outcome and exposure. Confounding variables included sociodemographic

variables consisting of place of residence (rural/urban), and ecological belt (mountain/hill/terai), child-related variable consisting of sex of the child (male/female), and parents-related variables consisting of parity (Primipara/Multipara), mother's anemia status (a mother was considered anemic if the level was less than 11.0 gm/deciliter. [13]) and father's education (no education/basic level/secondary and higher level).

Statistical analysis

We conducted pre-analytical processing and statistical analysis using R version 4.3.2 [33] and R studio [34]. We carried out weighted descriptive and inferential analysis using the "survey" package to address complex survey design and non-response. ~~We presented p~~Parametric numerical variables ~~were presented~~ as mean (standard deviation) and non-parametric numerical variables as median (interquartile range). ~~with their 95% confidence interval (CI).~~ We presented categorical variables as frequency, percent, and their ~~95% confidence interval (CI)~~~~95% CI~~. We computed the prevalence and 95% CI using the Wilson method.

We performed multivariable multinomial logistic regression, to determine the association between ~~the coexistence of~~ ~~eo-existence of stunting, wasting, and anemia (normal/undernutrition only/ anemia only/ eo-existence)~~ undernutrition and anemia (none/undernutrition only/anemia only/coexistence) and ~~the~~ exposure variables listed in table 1, adjusting for potential confounding variables (place of residence, ecological belt, age of the child, sex of the child, mother's parity, father's education). We performed binary logistic regression to determine the association between undernutrition or anemia with the exposure variables. We checked for multicollinearity using variance inflation factor (VIF) and removed province (VIF= 13.31, greater than the cutoff of 10) from the regression models, as the information provided by the province ~~variable~~ was contained by the region and urban status variables. All regression analyses were conducted using complete cases after excluding observations with missing data. We calculated and presented crude odds ratio (cOR) and adjusted odds ratio (aOR) and their 95% CI.

207 Results

208 Table 2 presents the sociodemographic characteristics of 6-59 months of children stratified by age. Of
 209 the total U-5 children, 48.5% were female. Age distribution showed 10.4% were 6-12 months old,
 210 45.8% were 13-36 months old, and 43.8% were 37-59 months old. Most children (59.1%) were from
 211 the Terai region, while the fewest (5.6%) were from the mountain region. The ratio of children from
 212 urban to rural areas was 2:1. By province, the majority were from Madhesh (26.4%), followed by Koshi
 213 (17.9%), Lumbini (16.7%), and Bagmati (15.4%).

214 **Table 2: Characteristics of 6-59 months children**

Characteristics	Overall, %	6-12 months children, %	13 to 36 months children, %	37-59 months children, %
Total unweighted frequency, <i>n</i>	2395	256	1092	1047
Total weighted frequency, <i>n</i>	2335	244	1069	1022
Age of children in months, %		10.4	45.8	43.8
Sex, %				
Male	51.5	52.6	51.4	51.3
Female	48.5	47.4	48.6	48.7
Ecological region, %				
Mountain	5.6	5.6	5.6	5.6
Hill	35.3	41.7	34.4	34.7
Terai	59.1	52.7	60	59.8
Type of place of residence, %				
Urban	64.3	63.1	64.2	64.7
Rural	35.7	36.9	35.8	35.3
Province, %				
Koshi	17.9	18.7	17.2	18.5
Madhesh	26.4	20.7	27.2	27
Bagmati	15.4	20	15.8	13.9
Gandaki	7.2	6.7	6.1	8.5
Lumbini	16.7	15.6	17.1	16.6
Karnali	7.7	9.8	7.1	7.8
Sudurpashchim	8.6	8.5	9.6	7.7
Wealth quintile, %				
Poorest	24.5	22.2	24.4	25
Poorer	22	19.9	23.5	20.9
Middle	20.7	19.6	19.6	22
Richer	18.1	19.1	19.2	16.8
Richest	14.7	19.2	13.3	15.2
Mother's education, %				
No education	22.2	15.3	23.1	22.9
Basic	35.9	32.5	35.7	37
Secondary and higher	41.9	52.2	41.2	40

Missing, n	55	0	14	40
Father's education, %				
No education	10.7	6.1	10.8	11.8
Basic	41	42.2	41.1	40.7
Secondary and higher	48.3	51.7	48.1	47.6
Missing, n	322	27	135	160
Mother's age at delivery, %				
<20 years	19.8	18.3	19.2	20.8
20-34 years	76.3	78.9	77.7	74.2
35-49 years	3.9	2.8	3.1	5
Missing, n	259	21	104	134
Parity, %				
Primipara	32.1	41.3	37.3	24.2
Multipara	67.9	58.7	62.7	75.8
Missing, n	259	21	104	134
Mother's BMI, %				
Underweight (<18.5 kg/m ²)	14.6	12.2	17.8	11.9
Normal (18.5 to 24.9 kg/m ²)	61.7	58.1	62.2	62.1
Overweight or obese (≥25 kg/m ²)	23.7	29.7	20	26
Missing, n	399	22	185	192
Mother's participation in household decision-making, %				
No participation	33.5	32.5	34.6	32.6
Participation	66.5	67.5	65.4	67.4

n: weighted frequency; %: weighted percentage

Missing are presented as numbers

All percentages are column percentage

Figure 1 presents the prevalence of stunting, wasting and underweight among Nepalese children aged 6-59 months. The prevalence of stunting, wasting, and underweight were 25.9% (95% CI: 23.6 to 28.3), 7.8% (95% CI: 6.5 to 9.3), and 19.8% (95% CI: 17.6 to 22.1), respectively. Of the total children, 3.2% (95% CI: 2.5 to 4.3) had all three conditions, 13.5% (95% CI: 11.9 to 15.3) had two of the three conditions, and 16.8% (95% CI: 15.2 to 18.5) had one of the three conditions.

[Figure 1 here]

Figure 1: Prevalence of co-existence of different forms of undernutrition among A) children aged 6-12 months, B) children aged 13-36 months, C) children aged 36-59 months and D) all children aged 6-59 months.

231 The prevalence of anemia was 43.4% (95% CI: 40.9 to 45.9), which included 18.8% (95% CI:16.8 to
232 20.9) with moderate to severe anemia and 24.6% (95%CI: 22.7 to 26.6) with mild anemia. The
233 prevalence of moderate to severe anemia was highest in late infants, accounting for 39.9% (95% CI:
234 33.2 to 47.1) and decreased to 8.4% (95% CI: 6.7 to 10.8) in preschools (**Figure 2**).

235

236 **[Figure 2 here]**

237 *Figure 2: Prevalence of anemia among A) children aged 6-12 months, B) children aged 13-36*
238 *months, C) children aged 36-59 months and D) all children aged 6 – 59 months.*

239

240

241 The prevalence of co-existence of stunting-anemia, wasting-anemia, underweight-anemia,
242 undernutrition-anemia were 12.5% (95%CI: 10.9 to 14.3), 4.0% (95%CI: 2.9 to 4.9), 9.7% (95% CI:
243 8.7 to 11.7), and 16.0% (95% CI: 14.2 to18.0) respectively (**Figure 3A-D**). Of total children, 1.9%
244 (95%CI: 1.3 to 2.6) had all three undernutrition conditions and anemia, 6.4% (95%CI: 5.3 to 7.8) had
245 two undernutrition conditions and anemia, and 7.6% (95%CI: 6.6 to 8.9) had one undernutrition
246 condition and anemia. (*not presented in tables*)

247

248 **[Figure 3 here]**

249 *Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and*
250 *overall undernutrition*

Table 3 presents the prevalence of co-existence of anemia and undernutrition by age group and sex of child. The prevalence of co-existence of undernutrition and anemia was highest in children 13-36 months (overall: 22.0%, male: 20.7%, female: 23.7%) followed by 6-12 months old children (overall: 16.2%, male: 17.2%, female: 14.9%). Prevalence of only anemia was highest in 6-12 months (overall: 54.9%, male: 50.2%, female: 60.1%) and only undernutrition in 37-59 months old children (overall: 23.5% male: 21.0%, female: 26.2%).

Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

Characteristic	6-59 months children, % (95%CI)	6-12 months children, % (95%CI)	13-36 months children, % (95%CI)	37-59 months children, % (95%CI)
All children				
None	39.1 (36.6 to 41.7)	25.0 (19.4 to 31.6)	33.6 (30.5 to 36.9)	48.2 (44.3 to 52.1)
Only Anemia	27.4 (25.3 to 29.6)	54.9 (48.3 to 61.3)	29.5 (26.4 to 32.7)	18.6 (15.8 to 21.8)
Only undernutrition	17.5 (15.6 to 19.6)	3.9 (1.99 to 7.49)	14.9 (12.5 to 17.6)	23.5 (20.6 to 26.8)
Co-existence of undernutrition and anemia	16.0 (14.1 to 18.0)	16.2 (11.7 to 22.0)	22.0 (19.1 to 25.3)	9.6 (7.89 to 11.7)
Male children				
None	39.5 (36.0 to 43.1)	29.5 (21.0 to 39.6)	33.5 (29.0 to 38.4)	48.2 (43.0 to 53.5)
Only Anemia	27.9 (25.1 to 30.9)	50.2 (41.1 to 59.2)	30.7 (26.6 to 35.1)	19.6 (15.9 to 23.9)
Only undernutrition	16.5 (14.2 to 19.2)	2.9 (0.94 to 8.83)	15.4 (12.3 to 19.2)	21.0 (17.2 to 25.5)
Co-existence of undernutrition and anemia	16.1 (13.7 to 18.8)	17.4 (11.3 to 25.9)	20.4 (16.9 to 24.5)	11.2 (8.71 to 14.2)
Female children				
None	38.7 (35.4 to 42.1)	20.1 (13.3 to 29.3)	33.8 (29.5 to 38.4)	48.1 (43.0 to 53.3)
Only Anemia	26.8 (23.8 to 30.1)	60.1 (49.2 to 70.0)	28.2 (23.9 to 33.1)	17.7 (13.8 to 22.3)
Only undernutrition	18.6 (15.9 to 21.5)	4.9 (2.09 to 11.2)	14.3 (11.2 to 18.1)	26.2 (21.7 to 31.2)
Co-existence of undernutrition and anemia	15.9 (13.4 to 18.7)	14.9 (9.01 to 23.5)	23.7 (19.3 to 28.7)	8.0 (5.72 to 11.1)

%, weighted percent; CI: confidence interval; All percentages in the table are row percentage

Figure 4 and Table 4 present the distribution of anemia, undernutrition and co-existence across categories of quintile wealth, mother's education, household decision-making and mother's exposure to health programs on TV and radio. The prevalence of undernutrition and co-existence was highest in poorest wealth quintile and lowest in richest wealth quintile, and highest in children whose mother had no education.

[Figure 4 here]

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables (vertical bars represent 95% CIs)

Table 4: Distribution of undernutrition, anemia and their co-existence across exposure variables

Characteristics	6-12 months children*			13-36 months children*			37-59 months children*		
	Anemia only, %	Undernutrition only, %	Co-existence, %	Anemia only, %	Undernutrition only, %	Co-existence, %	Anemia only, %	Undernutrition only, %	Co-existence, %
Wealth quintile									
Poorest	55.9	4.3	19.0	24.2	19.7	26.6	12.3	30.9	13.2
Poorer	51.7	1.8	15.2	30.3	16.4	25.4	17.6	27.4	11.0
Middle	60.9	3.3	12.2	31.0	12.3	21.3	24.4	27.3	10.1
Richer	56.9	4.0	24.0	35.2	12.5	18.5	29.5	13.0	5.9
Richest	48.8	6.2	10.4	27.2	10.4	13.7	10.2	12.4	5.3
Mother's education									
No education	51.7	8.1	20.9	27.5	15.1	35.5	21.9	29.0	15.6
Basic	51.1	1.1	22.4	27.7	16.1	22.9	17.7	27.9	9.4
Secondary and higher	58.1	4.4	11.0	32.3	13.8	13.9	18.0	15.9	7.2
Participation in household decision-making	52.9	4.7	14.9	28.5	14.9	21.3	17.9	22.2	9.4
Exposure to health programs on TV/radio	64.3	4.7	11.9	24.6	15.5	14.2	16.6	26.7	10.0
Mother's BMI									
Underweight	50.3	6.9	25.3	22.1	22.5	33.4	28.3	22.5	13.2
Normal	58.0	2.9	17.1	30.3	14.4	21.1	15.7	26.2	10.8
Overweight or obese	46.1	5.0	13.9	37.7	10.2	12.9	22.1	18.8	5.5

%, weighted percentage

* Row percentage

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Table 5 presents the factors associated with undernutrition and anemia among children aged 6-59 months. In univariable regression, undernutrition was significantly associated with wealth quintile, mother's education, age of child, and maternal nutritional status. In multivariable regression,

undernutrition was found to be associated with age of child, wealth quintile, mother's education, and maternal nutrition status after adjusting for confounding variables. The odds of having undernutrition were 1.96 times among 13-36 months children and 1.70 times in the 37-59 months children compared to 6-12 months children. The odds of undernutrition were 44% (aOR: 0.56, 95%CI: 0.34 to 0.92) lower in children from the richer wealth quintile compared to the poorest, 36% (aOR: 0.64, 95%CI: 0.45 to 0.93) lower in the child whose mother had secondary or higher-level education. We did ~~not~~ find a significant association between undernutrition and mother participation in household decision-making[#] and exposure to health programs on TV/radio.

Table 5: Factors associated with undernutrition among children aged 6-59 months

Characteristic	Presence of undernutrition, n (%) *	Unadjusted models			Adjusted model [#]		
		cOR	95% CI	p-value	aOR	95% CI	p-value
Age of child							
6-12 months	49 (20.1)	Ref			Ref		
13-36 months	394 (36.9)	2.32	1.67; to 3.22	<0.001	1.96	1.33; to 2.90	<0.001
37-59 months	339 (33.2)	1.97	1.39; to 2.81	<0.001	1.70	1.12; to 2.59	0.013
Wealth quintile							
Poorest	246 (43.1)	Ref			Ref		
Poorer	195 (38.1)	0.81	0.62; to 1.06	0.122	0.93	0.67; to 1.30	0.680
Middle	162 (33.6)	0.67	0.50; to 0.88	0.005	0.78	0.54; to 1.13	0.194
Richer	109 (25.7)	0.46	0.32; to 0.66	<0.001	0.53	0.34; to 0.82	0.005
Richest	70 (20.2)	0.33	0.22; to 0.50	<0.001	0.56	0.34; to 0.92	0.024
Mother's education							
No education	234 (46.3)	Ref			Ref		
Basic	301 (36.7)	0.67	0.52; to 0.88	0.003	0.87	0.65; to 1.16	0.332
Secondary and higher	231 (24.2)	0.37	0.28; to 0.49	<0.001	0.64	0.45; to 0.93	0.019
Mother's Participation in household decision making							
No participation	279 (35.7)	Ref			Ref		
Participation	503 (32.4)	0.86	0.70; to 1.07	0.173	0.82	0.63; to 1.06	0.133
Exposure to health programs on TV/radio							
No	615 (34.5)	Ref			Ref		
Yes	145 (31.0)	0.85	0.67; to 1.09	0.199	0.92	0.69; to 1.24	0.595
Mother's BMI							
Normal	411 (34.5)						
Underweight	132 (46.6)	1.66	1.23; to 2.23	<0.001	1.48	1.09; to 2.01	0.013
Overweight or obese	106 (23.1)	0.57	0.43; to 0.75	<0.001	0.75	0.56; to 1.01	0.057

n: weighted frequency; %: weighted percent; Ref: reference group; cOR: crude odds ratio; aOR: adjusted odds ratio; CI: confidence interval; * Row percent

[#] adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

Table 6 presents the factors associated with anemia among children aged 6-59 months. The factors associated were age of the child, wealth quintile, and anemia status of the mother. The odds of having anemia was 41% (aOR: 0.59, 95%CI: 0.36 to 0.97) lower among children belonging to richest quintile compared to poorest. The odds of having anemia was 64% (aOR: 0.36, 95%CI: 0.24 to 0.53) lower

298 among children aged 13-36 months and 87% (aOR: 0.13; 95%CI: 0.09 to 0.20) lower child aged 37-59
299 months compared to child aged 6-12 months after adjusting for potential confounding variables.
300

301 **Table 6: Factors associated with anemia among children aged 6-59 months**

Characteristic	Prevalence of anemia, n (%) *	Unadjusted models			Adjusted model [#]		
		cOR	95% CI	p-value	aOR	95% CI	p-value
Age of child							
6-12 months	173 (71.1)	Ref			Ref		
13-36 months	551 (51.5)	0.43	0.31;to 0.60	<0.001	0.36	0.24;to 0.53	<0.001
37-59 months	289 (28.3)	0.16	0.11;to 0.23	<0.001	0.13	0.09;to 0.20	<0.001
Wealth quintile							
Poorest	239 (41.8)	Ref			Ref		
Poorer	233 (45.5)	1.16	0.88;to 1.53	0.291	0.88	0.63;to 1.23	0.451
Middle	222 (46.0)	1.19	0.90;to 1.56	0.215	0.89	0.61;to 1.30	0.531
Richer	209 (49.2)	1.35	1.02;to 1.80	0.038	1.07	0.72;to 1.60	0.725
Richest	110 (31.9)	0.65	0.46;to 0.93	0.02	0.59	0.36;to 0.97	0.037
Mother's education							
No education	265 (52.4)	Ref			Ref		
Basic	347 (42.4)	0.67	0.52;to 0.86	0.002	0.80	0.57;to 1.13	0.210
Secondary and Higher	388 (40.6)	0.62	0.48;to 0.80	<0.001	0.84	0.56;to 1.26	0.409
Mother's Participation in decision making							
No participation	365 (46.6)	Ref			Ref		
Participation	648 (41.7)	0.82	0.65;to 1.03	0.095	0.76	0.55;to 1.05	0.099
Exposure to health programs on TV/radio							
No	813 (45.6)	Ref			Ref		
Yes	179 (38.4)	0.74	0.59;to 0.94	0.012	1.00	0.74;to 1.35	0.990
Mother's BMI							
Underweight	149 (52.5)	1.46	1.07;to 1.98	0.016	1.20	0.84;to 1.71	0.319
Normal	516 (43.2)	Ref			Ref		
Overweight or obese	189 (41.1)	0.92	0.70;to 1.21	0.548	1.19	0.87;to 1.63	0.286

302 n: weighted frequency; %: weighted percent

303 * Row percent

304 Ref: reference group; cOR: crude odds ratio; aOR: adjusted odds ratio; CI: confidence interval

305 # adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth,
306 mother's anemia status, and adjusted for each other

309 Table 7 presents the factors associated with co-existence of undernutrition and anemia among children
310 aged 6-59 months. The co-existence of undernutrition and anemia was significantly associated with the
311 richest wealth quintile, the mother's education status, the mother's BMI, and presence of anemia in
312 children. The children from the richest wealth quintile were 53% (aOR: 0.47; 95%CI: 0.26 to 0.86) less
313 likely to have co-existence of undernutrition and anemia than children from the poorest wealth quintile.
314 The children whose mothers had secondary or higher-level education had 48% (aOR: 0.52; 95%CI:
315 0.32 to 0.86) lower odds of having co-existence compared to children whose mothers had no formal

education. The odds of having co-existence of undernutrition and anemia was 34% (aOR: 0.66; 95%CI: 0.47 to 0.94) lower in the children whose mother participate in household decision-making after adjusting for confounding variables. The child ~~with whose mother was~~ underweight ~~mother~~ was 1.80 times (95%CI: 1.20 to 2.70) more likely and mother with anemia were 2.43 times more likely to have co-existence compared to a child whose mother had normal nutrition status.

Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged 6-59 months

Exposure variables	Anemia only vs Normal			Undernutrition only vs Normal			Co-existence vs Normal		
	aOR [#]	95% CI	p-value	aOR [#]	95% CI	p-value	aOR [#]	95% CI	p-value
Age of child (ref: 6-12 months child)									
13-36 months	0.37	0.25 to 0.54	<0.001	3.02	1.32 to 6.91	0.009	0.68	0.42 to 1.11	0.123
37-59 months	0.16	0.11 to 0.24	<0.001	3.16	1.39 to 7.20	0.006	0.21	0.12 to 0.35	<0.001
Wealth Quintile (ref: Poorest)									
Poorer	0.94	0.63 to 1.40	0.77	0.98	0.65 to 1.48	0.928	0.8	0.51 to 1.25	0.325
Middle	0.98	0.64 to 1.49	0.917	0.83	0.53 to 1.30	0.412	0.61	0.37 to 1.01	0.054
Richer	1.04	0.67 to 1.61	0.869	0.48	0.28 to 0.81	0.006	0.58	0.34 to 0.99	0.048
Richest	0.49	0.30 to 0.79	0.004	0.38	0.21 to 0.67	0.001	0.47	0.26 to 0.86	0.015
Mother's education (ref: no education)									
Basic	0.87	0.60 to 1.27	0.473	1.00	0.67 to 1.49	0.996	0.70	0.46 to 1.06	0.094
Secondary and higher	1.02	0.67 to 1.56	0.923	0.87	0.54 to 1.38	0.543	0.52	0.32 to 0.86	0.010
Participation in HH decision-making (ref: nonparticipation)	0.69	0.51 to 0.93	0.016	0.73	0.52 to 1.02	0.064	0.66	0.47 to 0.94	0.02
Exposure to health programs on TV/radio (ref: No)	1.04	0.77 to 1.42	0.783	1.00	0.70 to 1.42	0.996	0.89	0.59 to 1.34	0.572
Mother's BMI (ref: normal weight)									
Underweight	1.17	0.80 to 1.71	0.426	1.44	0.96 to 2.16	0.075	1.80	1.20 to 2.70	0.005
Overweight and obese	1.30	0.97 to 1.74	0.076	0.89	0.63 to 1.28	0.539	0.88	0.59 to 1.32	0.529

Normal: participants having neither undernutrition nor anemia
ref: reference group; **aOR:** adjusted odds ratio; **CI:** confidence interval
Bold represents significance at 95% confidence level
[#] adjusted for ~~age~~, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, mother's anemia status, and adjusted for each other

Discussion:

Our analysis of a national survey revealed significant issues related to undernutrition, anemia and their co-existence among Nepalese children in 2022. Overall, the prevalence of undernutrition and anemia was 33.5% and 43.4%, respectively. The prevalence of co-existence between undernutrition and anemia was 16.0%. Specifically, the co-existence of stunting, wasting, and underweight with anemia was 12.5%, 4.0%, and 9.7%, respectively. Undernutrition demonstrated significant associations with age of child, household wealth, ~~maternal~~ mother's education, ~~parity~~, and ~~mother's~~ maternal BMI. Anemia was associated with both household wealth and maternal anemia status. The co-existence of undernutrition and anemia exhibited significant associations with wealth, maternal education, maternal involvement in household decision-making, parity, mother's ~~maternal~~ BMI, and the child's anemia status. However, none of the three outcomes undernutrition(undernutrition, anemia, and their co-occurrence) ~~-were not~~ found to be associated with maternal exposure to health programs on television and radio.

A study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and among children aged 6-59 months (57.3%) [35]. These figures are higher than the prevalence of undernutrition among 6-59 months children in Nepal (33.5%) found in our study. Our findings indicate that Nepal's burden, though lower, remains substantial. These differences may reflect variations in economic development, food security, and maternal and child health interventions across countries [36].

The prevalence of anemia among 6-59 months children in our study (43.5%) was higher than the global prevalence (39.8%) but lower than that reported among African children (60.2%) [12]. Our study also demonstrates a 9.1 percentage-point reduction in anemia prevalence between 2016 (52.6%) [37] and 2022, indicating progress in addressing this issue. Comparable rates have been observed in neighboring countries. A meta-analysis conducted by Kundu et al in Bangladesh reported a prevalence of 46.8% [38], which was similar to our finding. The latest National Family Health Survey (2019-21) in India reported a prevalence of 67.1%, which is higher than that observed in Nepal [39]. The decline in anemia prevalence in Nepal is encouraging but underscores the need for sustained efforts to address this condition, particularly among vulnerable populations [23].

A systematic review reported the pooled prevalence rates of wasting-anemia and stunting-anemia in least-developed countries to be 5.4% and 19.5%, respectively [40]. These rates are higher than our findings from Nepal, which were 4.0% and 12.5%, respectively.

The high prevalence of co-existence of undernutrition and anemia in our study (16%) suggests that affected children are ~~trapped in a vicious cycle~~ caught in an interlinked cycle of micronutrients deficiencies. Undernutrition impairs iron absorption and utilization, while anemia exacerbates undernutrition by reducing the body's ability to effectively utilize nutrients.

The children from the richest wealth quintile had lower odds of having the co-existence of anemia and undernutrition compared to those from the poorest quintile. This wealth-related equity gap in undernutrition among children aged 6–59 months is attributed to household economic status, which influences food insecurity and health care utilization when a child is ill [41–43].

Children from families in which mothers participated in household decision-making and mothers with secondary or higher education were less likely to experience the co-existence of anemia and undernutrition, as well as anemia alone, compared to children without these conditions. The health and nutrition of the child largely depend on the mother, who is typically the primary caretaker and responsible for awareness of the child's health. Analyses of DHS data from Nepal and India suggest that women's decision-making authority (maternal autonomy in household decision-making) improves the children's nutritional status and reduces child mortality, even after controlling for education and wealth [44].

Strengths and Limitations

This study has several strengths. First, it utilized complex survey analysis to account for survey design and non-response rates. Second, the findings are generalizable to children aged 6–59 months in Nepal and to children of the same age group in demographically and economically similar countries, as the study used a nationally representative dataset.

This study ~~is not free from~~has several limitations. First, due to the missing data, we could not include some of the important dietary factors like minimum dietary diversity score and child's birth weight. Second, because of the cross-sectional nature of the study, the directionality of associations cannot be established. Third, the results from this study may not be generalizable to populations outside Nepal with different demographic and economic status. Lastly, the wide confidence intervals observed in the associations between children's nutritional status and anemia may suggest limited statistical power, likely due to small sample sizes within certain subgroups.

Conclusion

The prevalence of undernutrition, anemia, and their co-existence remains high among children aged 6–59 months in Nepal. The co-existence of anemia and undernutrition was less likely among children from the highest wealth quintile, those whose mothers attained at least secondary level education, and those whose mothers participated in household decision-making. In contrast, co-existence was more ~~prevalent-likely~~ among children whose mothers had poor maternal nutritional status. It is essential to implement holistic programs and targeted interventions to empower poor families, improve

396 mother's~~external~~ education, strengthen women's roles in household-~~decision-~~making, and ultimately
397 improve the children's health.

398

399 **Data and analysis code availability**

400 We used publicly available data from Nepal Demographic and health survey database upon request
401 from the official website of "The DHS program". The dataset is available at:
402 https://dhsprogram.com/data/dataset/Nepal_Standard-DHS_2022.cfm?flag=1

403 The analysis code and data used for this analysis is available on GitHub:
404 <https://github.com/Adrisyaone/Undernutritiona-and-anemia>

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407 **Conflict of interest**

408 The authors declare no conflicts of interest.

409

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554 **Legends of tables and figures**

555 **Tables**

556 Table 1: List of exposure variables with their definition

557 Table 2: Characteristics of 6-59 months children

558 Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

559 Table 4: Distribution of undernutrition, anemia and their co-existence across exposure variables

560 Table 5: Factors associated with undernutrition among children aged 6-59 months

561 Table 6: Factors associated with anemia among children aged 6-59 months

562 Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged 6-59
563 months

564

565 **Figures**

566 Figure 1: Prevalence of co-existence of different forms of undernutrition among A) children aged 6-12
567 months, B) children aged 13-36 months, C) children aged 37-59 months, and D) all children aged 6-59
568 ~~months-months.~~

569 Figure 2: Prevalence of anemia among A) children aged 6-12 months, B) children aged 13-36 months,
570 C) children aged 37-59 months, and D) all children aged 6-59 months.

571 Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall
572 undernutrition

573 Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across
574 exposure variables